

# McCann model for rhombohedral graphene

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# Chapter 1

## Anomalous Hall Effect

### 1.1 Introduction

This report analyzes the behavior of ABC-stacked multilayer graphene using the McCann model. We specifically compare the cases for  $N = 1$  (monolayer) and  $N = 5$  layers, focusing on their energy bands, Berry curvature, and the convergence of the local Chern number contribution.

### 1.2 Model and Analytical Equations

The system is described by a 2x2 effective Hamiltonian in the vicinity of a K point (low energy):

$$H = \begin{pmatrix} \Delta & X(\mathbf{k}) \\ X^\dagger(\mathbf{k}) & -\Delta \end{pmatrix} \quad (1.1)$$

where  $2\Delta$  is the energy gap and  $X(\mathbf{k})$  is the off-diagonal coupling term. In polar coordinates  $(k, \phi)$ , the term  $X$  for  $N$  layers is given by:

$$X(k, \phi) = \frac{(vke^{i\xi\phi})^N}{(-\gamma_1)^{N-1}} \quad (1.2)$$

where  $v = \frac{\sqrt{3}}{2}\gamma_0$  in natural units ( $a = 1, \hbar = 1$ ) and  $\xi = \pm 1$  is the valley index.

#### 1.2.1 Energy Bands

The eigenvalues of the Hamiltonian represent the energy bands:

$$\varepsilon(\mathbf{k}) = \pm E(\mathbf{k}) = \pm \sqrt{|X(\mathbf{k})|^2 + \Delta^2} \quad (1.3)$$

#### 1.2.2 Eigenstates

The normalized eigenstates for the positive and negative bands are given by:

$$|\psi_+\rangle = \frac{1}{\mathcal{N}_+} \begin{pmatrix} E + \Delta \\ X^\dagger \end{pmatrix}, \quad |\psi_-\rangle = \frac{1}{\mathcal{N}_-} \begin{pmatrix} -X \\ E + \Delta \end{pmatrix} \quad (1.4)$$

where  $\mathcal{N}_\pm$  are the normalization constants.

### 1.2.3 Berry Curvature

The Berry curvature  $\Omega$  is calculated using the Kubo-like formula. For the results presented in this report, we specifically show the Berry curvature calculated for the positive energy band:

$$\Omega_n = i \sum_{m \neq n} \frac{\langle \psi_n | \nabla_{\mathbf{k}} H | \psi_m \rangle \times \langle \psi_m | \nabla_{\mathbf{k}} H | \psi_n \rangle}{(E_n - E_m)^2} \quad (1.5)$$

Here, the gradient is calculated in polar coordinates as:

$$\nabla_{\mathbf{k}} = \frac{\partial}{\partial k} \vec{e}_k + \frac{1}{k} \frac{\partial}{\partial \phi} \vec{e}_\phi \quad (1.6)$$

## 1.3 Numerical Results

### 1.3.1 Energy Bands Comparison

Figure 1.1 shows the energy bands for  $N = 1$  and  $N = 5$ . For  $N = 1$ , the dispersion is linear at high  $k$  (Dirac-like), while for  $N = 5$  it is much flatter near  $k = 0$  due to the  $k^N$  dependence.

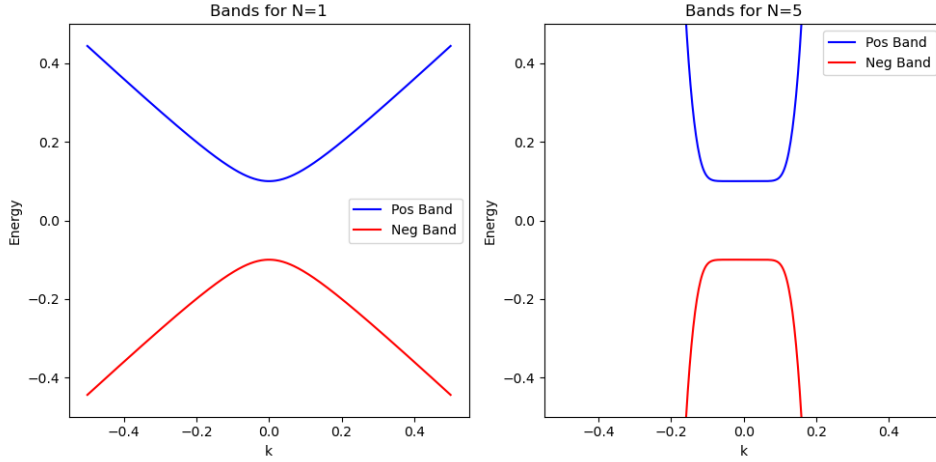


Figure 1.1: Energy bands for N=1 and N=5 layers.

### 1.3.2 Berry Curvature

The Berry curvature for  $N = 1$  peaks sharply at  $k = 0$ . However, for  $N = 5$ , the curvature is suppressed at the center and peaks at finite  $k$  (Figure 1.2).

## 1.4 Berry Curvature Integration

The local Chern number contribution is obtained by integrating the Berry curvature over a region in  $k$ -space. While the McCann model is isotropic and naturally expressed in polar coordinates, the numerical integration is performed over a uniform Cartesian meshgrid  $(k_x, k_y)$ . This choice provides several numerical advantages:

- **Uniform Sampling Density:** In a Cartesian grid, every point represents an identical area element  $\Delta A = \Delta k_x \Delta k_y$ . In a polar grid, the area element  $dA = k dk d\phi$  is proportional to  $k$  and vanishes as  $k \rightarrow 0$ . Since the Berry curvature is often highly concentrated near the Dirac point ( $k = 0$  for monolayer graphene), the Cartesian grid ensures high sampling density where it is most needed without requiring a non-linear radial distribution.

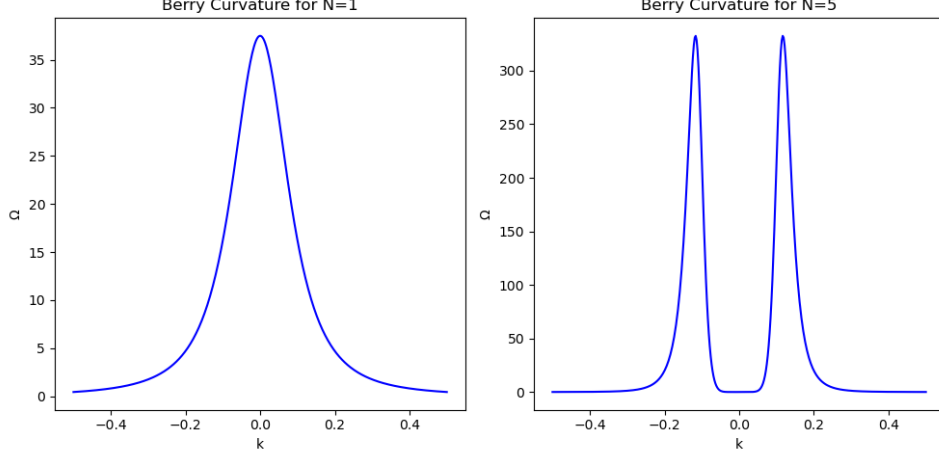


Figure 1.2: Berry curvature comparison (Positive Band).

- **Singularity Avoidance:** The Cartesian coordinate system avoids the coordinate singularity at the origin inherent to polar grids, which simplifies the handling of the peak in  $\Omega$  at  $k = 0$  for  $N = 1$ .

To ensure a consistent comparison between different layer numbers  $N$ , we define the integration domain using an energy cutoff  $E_{max}$  rather than a fixed momentum radius. The integral is evaluated numerically as:

$$C_{local}(E_{max}) = \frac{1}{2\pi} \iint_{E(\mathbf{k}) \leq E_{max}} \Omega_{pos}(\mathbf{k}) dk_x dk_y \approx \frac{\Delta A}{2\pi} \sum_{i \in \text{mask}} \Omega_{pos}(\mathbf{k}_i) \quad (1.7)$$

where the sum runs over all points  $k_i$  satisfying the energy condition  $E(\mathbf{k}_i) \leq E_{max}$ , and  $\Delta A$  is the constant area per grid point.

Figure 1.3 shows that as the energy cutoff  $E_{max}$  increases, the integral converges to  $N/2$ .

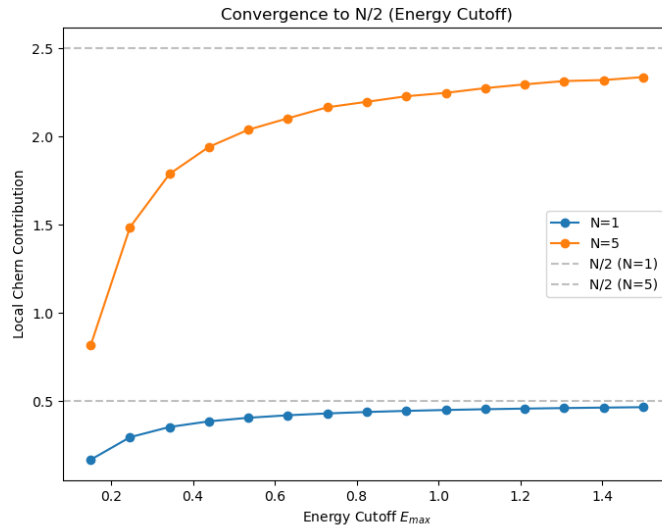


Figure 1.3: Convergence of the local Chern number contribution to  $N/2$  as a function of the energy cutoff  $E_{max}$ .

## 1.5 Valley Comparison

The two K-valleys in the hexagonal lattice, indexed by  $\xi = \pm 1$ , are related by time-reversal symmetry. While the energy dispersion is identical for both valleys, the Berry curvature of a given band changes its sign:

$$\begin{aligned}\mathcal{T} : \mathbf{k} &\rightarrow -\mathbf{k} \\ E(-\mathbf{k}) &= E(\mathbf{k}) \\ \Omega(-\mathbf{k}) &= -\Omega(\mathbf{k})\end{aligned}\tag{1.8}$$

Figure 1.4 compares the positive energy band and the corresponding Berry curvature for  $N = 5$  layers, showing the sign inversion between the two K valleys.

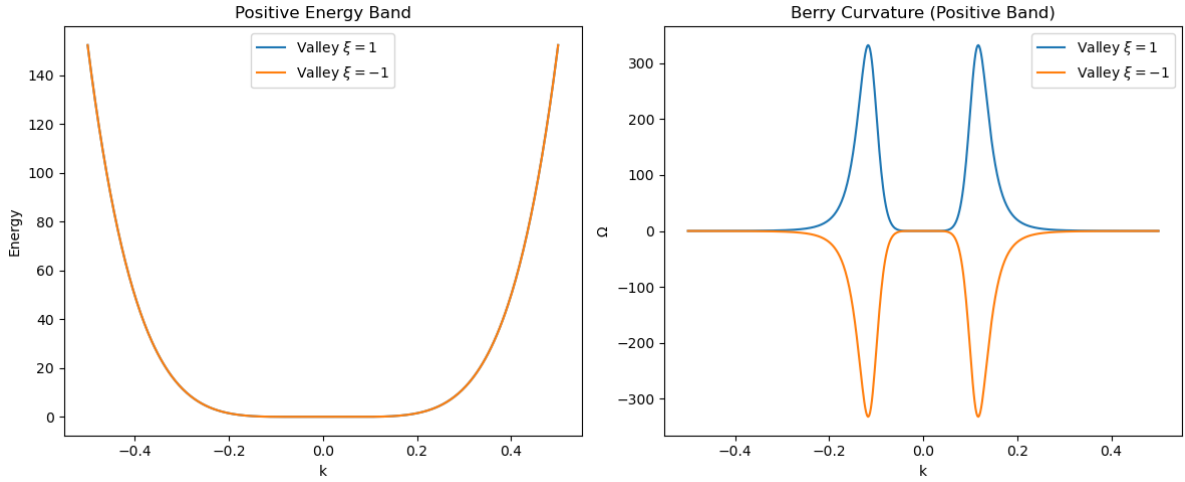


Figure 1.4: Comparison of the positive energy band (left) and Berry curvature (right) for valleys  $\xi = 1$  and  $\xi = -1$  ( $N = 5$  layers).

Consequently, the local Chern number contributions for  $N = 5$  have opposite signs. The integration of the Berry curvature over a circle of radius  $k=1$  gives:

- Valley  $\xi = 1$ :  $C_{local} = 2.499949$
- Valley  $\xi = -1$ :  $C_{local} = -2.499949$

The total Chern number for the system, considering both valleys, would vanish in the presence of time-reversal symmetry ( $C_{tot} = C_K + C_{K'} = 0$ ).

## 1.6 Anomalous Hall Effect with Spin

Recent additions to the project include the calculation of the Anomalous Hall Effect (AHE) considering both spin and valley degrees of freedom.

### 1.6.1 Magnetic AHE at $\mu = 0$

We calculate the total Hall conductivity  $\sigma_{xy}$  for the system at the Fermi level  $\mu = 0$ . We use  $N = 5$  layers and a temperature  $T = 20$  K. The gap parameters used are:

- Valley  $K$  ( $\xi = 1$ ):  $\Delta_{\uparrow} = 0.1$ ,  $\Delta_{\downarrow} = 0.0$
- Valley  $K'$  ( $\xi = -1$ ):  $\Delta_{\uparrow} = 0.0$ ,  $\Delta_{\downarrow} = -0.1$

The calculated total conductivity for the spin-valley case (without SOC) is:

$$\sigma_{xy}^{total}(\mu = 0) \approx -5.000 \, e^2/h \quad (1.9)$$

which corresponds to the sum of the Chern contributions of the occupied bands ( $N/2 = 2.5$  per valley per spin when applicable).

### 1.6.2 AHE vs. Chemical Potential

The dependence of the AHE on the chemical potential  $\mu$  was scanned using the following parameters:

- Valley  $K$ :  $\Delta_{\uparrow} = 0.1$ ,  $\Delta_{\downarrow} = 0.0$
- Valley  $K'$ :  $\Delta_{\uparrow} = 0.0$ ,  $\Delta_{\downarrow} = -0.2$

Figure 1.5 shows the results of the scan. It is important to note that the convergence to 0 far from  $\mu = 0$  (as well as the convergence to  $N/2$  when there is a plateau) is very slow and difficult to fully observe in numerical simulations due to the long tails of the Berry curvature.

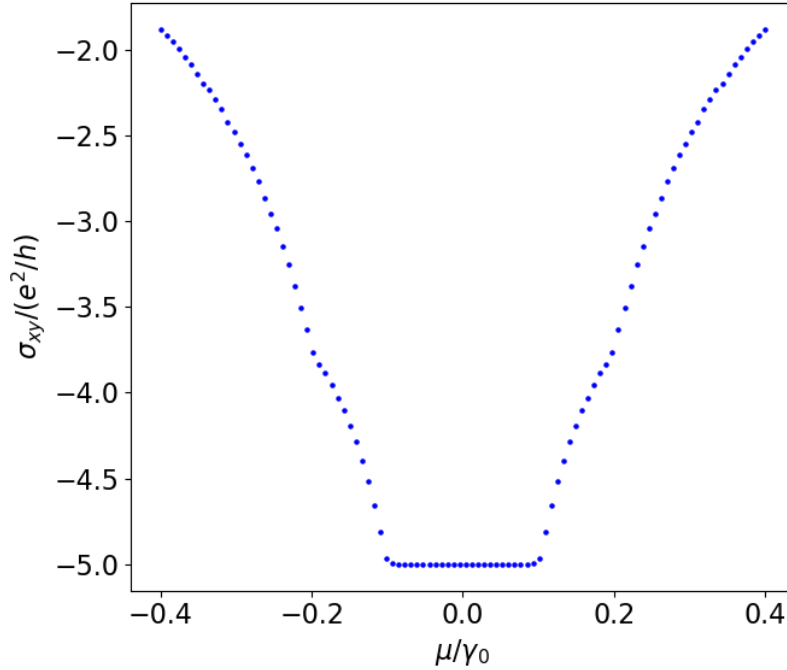


Figure 1.5: Total Hall conductivity  $\sigma_{xy}$  as a function of the chemical potential  $\mu/\gamma_0$  including spin and SOC.

## Chapter 2

# NonLinear AHE and positional shift

### 2.1 Hamiltonian with Trigonal Warping

In polar coordinates  $(k, \phi) \in ([0, +\infty[, [0, 2\pi])$ , we define  $\pi = vke^{i\xi\phi}$ .

$$\mathcal{H}_{\xi,s} = \begin{pmatrix} \Delta_{\xi,s} & X(\pi) \\ X^\dagger(\pi) & -\Delta_{\xi,s} \end{pmatrix} \quad (2.1)$$

To introduce the trigonal warping, we expand  $X(\pi)$  to first order in perturbations  $(v_3, \gamma_2)$  with  $N = n_1 + 2n_2 + 3n_3$ ,  $n_2$  being the order of  $v_3$  and  $n_3$  the order of  $\gamma_2$ .  $X_0$  is the unperturbed zero order term, then we write  $X_{n_2, n_3}$ :

$$X_0 = \frac{\pi^N}{(-\gamma_1)^{N-1}} \quad (2.2)$$

$$X_{10} = (N-1) \frac{v_3 \pi^{N-2} \pi^\dagger}{v(-\gamma_1)^{N-2}} = -\frac{4v_3 v^3}{\gamma_1^3} k^4 e^{i2\phi} \quad (2.3)$$

$$X_{01} = \frac{(N-2)}{2} \gamma_2 \left( \frac{\pi}{-\gamma_1} \right) = \frac{3v^2 \gamma_2}{2\gamma_1^2} k^2 e^{i2\phi} \quad (2.4)$$

### 2.2 Nonlinear Conductivity

For a given band:

$$\sigma^{ijl} = \frac{\partial j_i}{\partial E_j \partial E_l} = \frac{e^3}{\hbar^2} \int \frac{d^2 k}{(2\pi)^2} \frac{\partial f_n}{\partial \varepsilon_n} \left[ 2v_i G_{jl} - \frac{1}{2} (v_j G_{li} + v_l G_{ji}) \right] \quad (2.5)$$

With the quantum metric for band 0 defined by:

$$G_{ij} = 2\text{Re} \sum_{n \neq 0} \frac{(V_i)_{0n} (V_j)_{n0}}{(\varepsilon_0 - \varepsilon_n)^3} \quad (2.6)$$

### 2.3 Symmetry Constraints on the Conductivity

- **Mirror  $\mathcal{M}_y$  ( $y \rightarrow -y$ ):** components with an odd number of  $y$  are 0.
- **$C_{3z}$  rotation:** only non-zero components are related:  $\sigma_{xxx} = -\sigma_{xyy} = -\sigma_{yyx} = -\sigma_{yxx}$

## 2.4 Evolution of Nonlinear Conductivity

In this chapter, we analyze the behavior of the nonlinear conductivity  $\sigma_{xxx}$  related to the positional shift. We focus on its dependence on the chemical potential  $\mu$  for various gap configurations defined by the parameters  $\Delta_{v,s}$ , where  $v \in \{K, K'\}$  and  $s \in \{\uparrow, \downarrow\}$ .

The plots below show the evolution of  $\sigma_{xxx}$  for different scenarios of gap symmetries. The shaded regions represent the energy gaps for each spin and valley.

### 2.4.1 Scenario A: Symmetric Gaps, Zero Minors

Parameters:  $\Delta_{K\uparrow} = 0.1, \Delta_{K\downarrow} = 0.0, \Delta_{K'\uparrow} = 0.0, \Delta_{K'\downarrow} = -0.1$ . Here  $\Delta_{1\uparrow} = -\Delta_{2\downarrow}$  and  $\Delta_{1\downarrow} = \Delta_{2\uparrow} = 0$ .

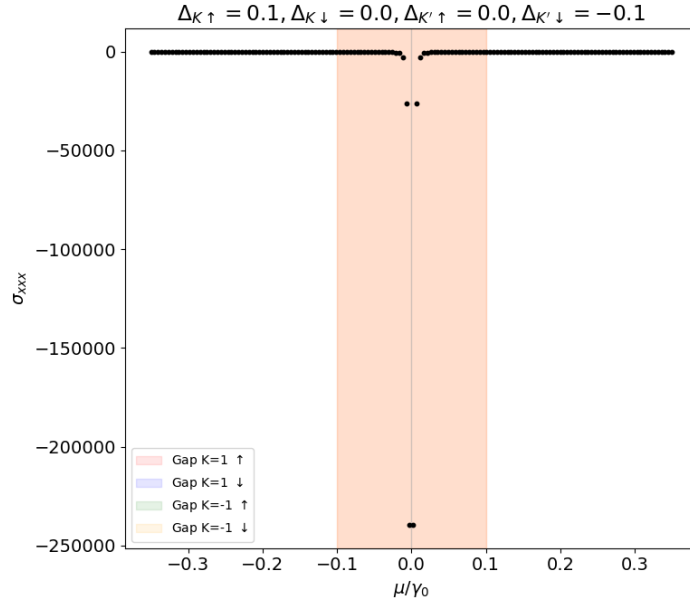


Figure 2.1: Nonlinear conductivity for symmetric gaps with zero minor gaps.

### 2.4.2 Scenario B: Symmetric Gaps, Non-Zero Minors

Parameters:  $\Delta_{K\uparrow} = 0.1, \Delta_{K\downarrow} = 0.05, \Delta_{K'\uparrow} = 0.05, \Delta_{K'\downarrow} = -0.1$ . Here  $\Delta_{1\uparrow} = -\Delta_{2\downarrow}$  and  $\Delta_{1\downarrow} = \Delta_{2\uparrow} \neq 0$ .

### 2.4.3 Scenario C: Asymmetric Gaps, Zero Minors

Parameters:  $\Delta_{K\uparrow} = 0.1, \Delta_{K\downarrow} = 0.0, \Delta_{K'\uparrow} = 0.0, \Delta_{K'\downarrow} = -0.2$ . Here  $\Delta_{1\uparrow} \neq -\Delta_{2\downarrow}$  and  $\Delta_{1\downarrow} = \Delta_{2\uparrow} = 0$ .

### 2.4.4 Scenario D: Asymmetric Gaps, Non-Zero Minors

Parameters:  $\Delta_{K\uparrow} = 0.1, \Delta_{K\downarrow} = 0.05, \Delta_{K'\uparrow} = 0.05, \Delta_{K'\downarrow} = -0.2$ . Here  $\Delta_{1\uparrow} \neq -\Delta_{2\downarrow}$  and  $\Delta_{1\downarrow} = \Delta_{2\uparrow} \neq 0$ .



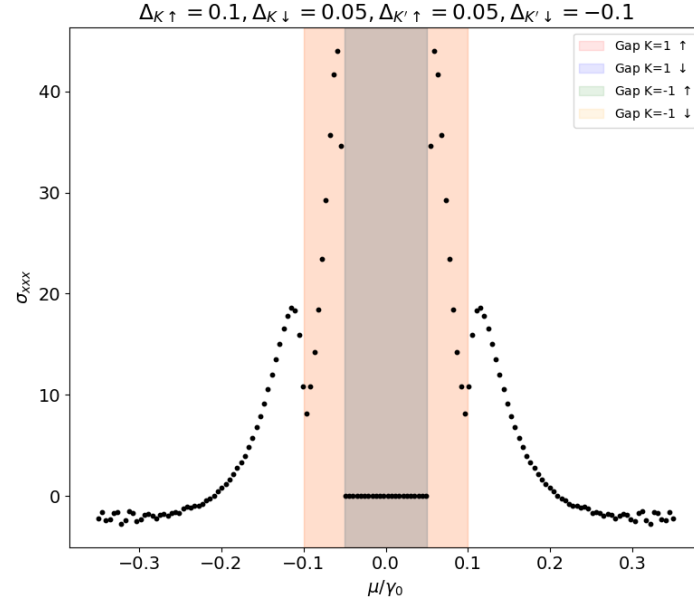


Figure 2.2: Nonlinear conductivity for symmetric gaps with non-zero minor gaps.

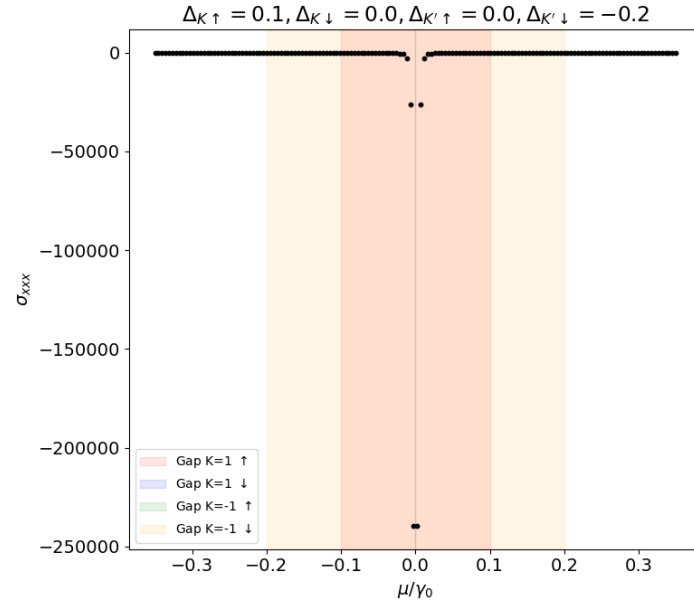


Figure 2.3: Nonlinear conductivity for asymmetric gaps with zero minor gaps.

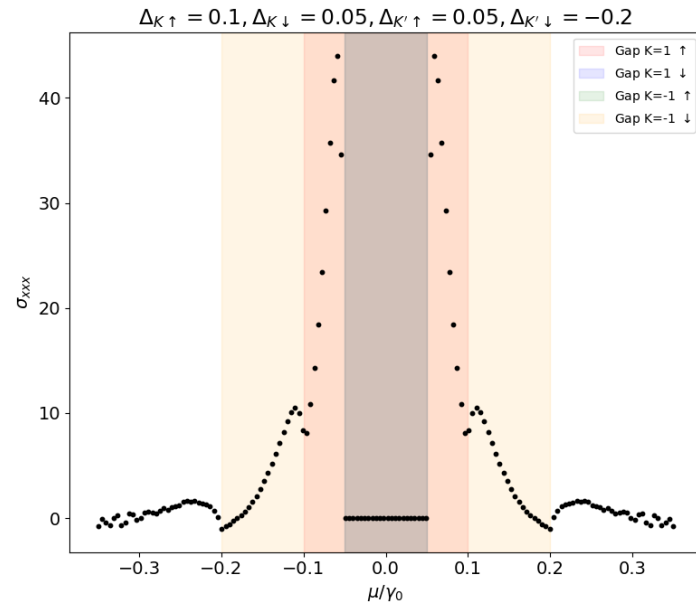


Figure 2.4: Nonlinear conductivity for asymmetric gaps with non-zero minor gaps.